Lettuce greenhouse production control using Reinforcement Learning

Project of Advanced Machine Learning course.

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Introduction

In this project, you will investigate reinforcement learning algorithms to control the indoor climate of a lettuce greenhouse in order to maximise net profit.

Greenhouses are a crucial protected horticulture system that provides fresh food for the growing global population. However, it is also a significant resource consumer, such as the large amount of energy usage from LED lighting and heating to maintain an ideal growing climate for the crops. (Graamans *et al.*, 2018). Therefore, the main objective of greenhouse production control is realising resource-effective crop growth and development through operating indoor climate actuators (e.g. lighting, heating, CO_2 dosing, ventilation, screening, watering). Modern sensing, control and computing systems are equipped in high-tech greenhouses, but the operating set-points for the climate actuators still rely on growers. This management strategy can ensure a greenhouse climate at certain levels suitable for crop growth without exceeding thresholds predefined according to growers' experience. However, as the production process is complex, including many unclear interactions, non-linearity, differences in response times, variability amongst individual plants, species and growth stages, as well as the uncertainty of outdoor climate, the human-perception-based control tends to be limited to guarantee optimal resource effective growing environment for the crops (Van Henten, 2013). Moreover, the number of experienced growers is rare, while the scale of the greenhouse production system is expanding worldwide. Therefore, using advanced techniques to autonomously control the greenhouse production system with more optimality reliably and robustly is in need with a significant challenge.

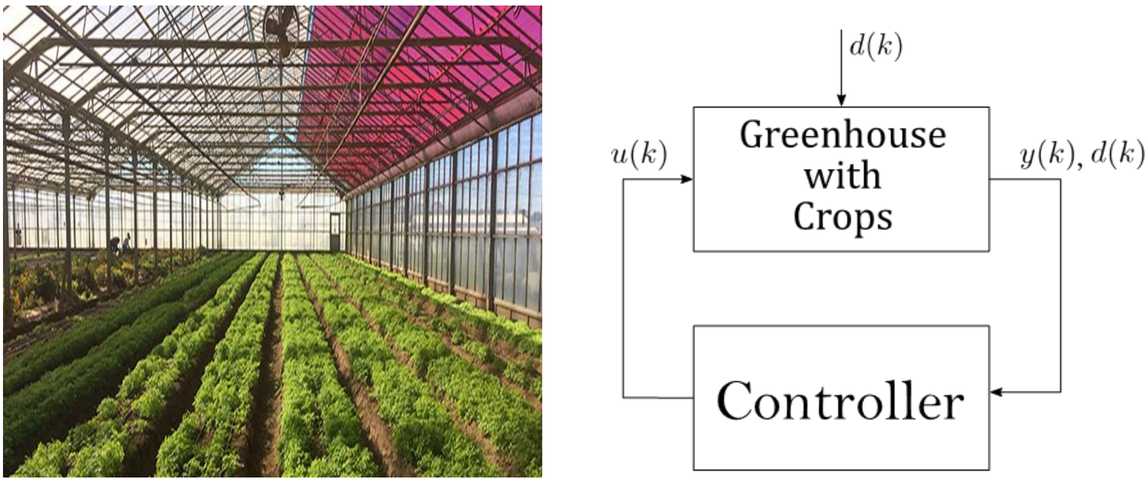

Figure 1: A lettuce greenhouse (left) and the basic control scheme (right)

The motivation of this project is to develop a climate control method for lettuce greenhouse using reinforcement learning (RL), aiming to optimise the production system with the desired growing climate, efficient resource usage, and, at the same time, adaptability to variable conditions (amongst individual plants, species and growing stages). The fast development of sensor technology in crops (e.g. leaf area, canopy transpiration, stomatal conductance, chlorophyll fluorescence) and climate (e.g. temperature, humidity, CO_2) makes RL application into greenhouse production possible.

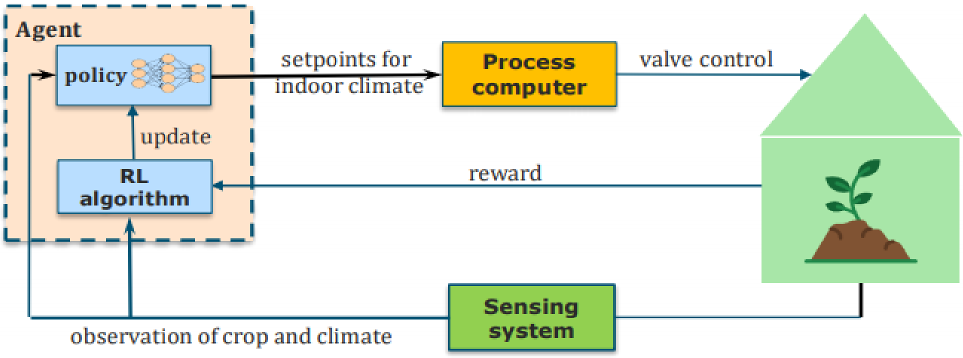

Figure 2: Schematic of greenhouse production control using RL (reinforcement learning) approach

RL can cover model-based and model-free types. In this project, the control objective will mainly focus on optimising resource usage for heating and carbon dioxide while satisfying growing requirements. Lettuce will be considered the crop in the greenhouse because of its short growing time, which will enable rapid data collection and the existing literature with a simple canopy to facilitate imaging and monitoring. Moreover, its growth dynamics are relatively simple, making it an ideal crop to model for simulation.

The assignment

This project aims to investigate different reinforcement learning methods to control the indoor climate of a lettuce greenhouse to achieve maximal net profit. In the Advanced Machine Learning course,
you learned different reinforcement learning methods that you could use to deal with the challenges mentioned above, such as deep reinforcement learning, Q-learning, SARSA, tabular methods, temporal difference learning, etc. Your task is to reason about the problem and investigate methods that you think are relevant to solving the assignment.

We have already modelled some lettuce greenhouse production system functionalities to get you started. The model of the greenhouse is derived from **(Boersma & van Mourik, 2021)**. The pdf of this paper can be found in the project folder.

Moreover, this folder contains an aml_project/ folder, which contains code for solving the lettuce greenhouse problem.

It consists of three folders:

- **common/**
  - helper_functions.py (functions used by the environment to simulate indoor greenhouse climate and crop growth)
  - plot_gh_variables.py (helper function to plot climate and crop trajectories)
- **environments/**
  - weather/ (folder with weather data required for greenhouse simulations)
  - lettuce_greenhouse.py (script that holds python class to simulate the environment)

- **experiments/**
    - example.ipynb (example notebook, how to simulate the greenhouse model using the environment wrapper)

The model consists of **four continuous state variables** (*x*):

- Lettuce dry weight [kg/m^2]
- Indoor CO_2 concentration [kg/m^3]
- Indoor air temperature [C]
- Indoor humidity [kg/m^3]

Four continuous weather variables (*d*):

- Outdoor radiation [W/m^2]
- Outdoor carbon dioxide concentration [kg/m^3]
- Outdoor air temperature [C]
- Outdoor humidity [kg/m^3]

and **three continuous control variables** (*u*):

- Supply rate of carbon dioxide [$mg/m^2/s$]
- Ventilation rate [mm/s]
- Energy supply by heating the system [W/m^2]

We also have **four continuous measurement variables**, which make the state variables more readable (*y*):

- Lettuce dry weight [g/m^2]
- Indoor CO_2 concentration [ppm]
- Indoor air temperature [C]
- Indoor relative humidity [%]

The following simulation parameters are given:

- Sampling period seconds (*h*)
- Number of seconds in the day (*c*)
- Number of days to simulation (*nDays*) [You could change this!]
- Total number of seconds in simulation (*L*)
- Number of timesteps in simulation (*N*)

Besides, we provide the following:
Action and observation spaces are also provided, but you can play with them.
Lower and upper bounds on observations and actions are provided.

Additional model parameters (*p*)
Initial state (x_0)
Also, weather measurements are loaded in for you.

In environment.py, we have implemented the following functions for you:

- *f(action, d)*
- *F(x, action, d, p)*
- *g()*

f() takes action and weather values for the right time as input and uses the Runge-Kutta 4 discretisation method to compute the state at the next step. *f()* makes use of *F()* to approximate the given differential equation. Function *g()* can transform the state variables into more readable measurement variables.

See the attached pdf of **Boersma & van Mourik's (2021)** paper for the exact model formulas.

Data

A yearly weather dataset is provided to sample weather patterns. You can include additional datasets to diversify and improve the training process.

Assignment Step 1:

Get familiar with the greenhouse climate-crop model.

Use the *example.ipynb* notebook to run simulations and test how the greenhouse model works and how you can interact with it. For example, how do you send control inputs and receive information from the environment?

Assignment Step 2:

Implement the *reward()* function. This is a crucial step as it depicts the signal your RL algorithm will learn from. If well-designed, your algorithm will learn a control policy that attains its desired objectives. A well-designed reward function should be minimally specified to give the learning algorithm maximum freedom to explore the state-action space to find the optimal yet realistic control policy.

Assignment Step 3:

Now, you can implement an RL agent that learns to control the lettuce greenhouse production system by regulating the indoor climate.

Remember that there are three control variables. You can choose to control all three, or you can select one or two that seem interesting to you and, for the other two, develop a rule-based controller.

You have the freedom to pick any RL algorithm you like. When making a choice, note that you must deal with a *continuous state-action space*.

You can use existing RL libraries for *Python* that contain ready-to-use RL algorithms. *For example, stable Baselines 3* is a well-established and actively managed library.

If you like a challenge, you can implement one yourself. Either using inheritance classes from the RL libraries or entirely from scratch.

Assignment Step 4:

You can start doing experiments When you have your RL agent running! You are free to devise any experiment you can think of. For example, test your algorithm's performance/robustness/generalisation/sensitivity. Also, compare it to the baseline policy that you developed earlier.

Assignment Step 5:

Task 2: Write a report.

Assessment of the project

The project will be assessed by the clarity and quality of your code (1/2) and the report (1/2). The
The report will be assessed on:

--- Clarity of text
--- Overall approach of comparing models
--- Explanation of methods
--- Motivation of methods/settings
--- Experiments and results
--- Presentation of results
--- Interpretation of results
--- Conclusion/Discussion

Note:

For additional inspiration, you can have a look at **(Morcego et al, 2023)**.

References

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